

Technical Datasheet Input / Output Modules with MQTT over Ethernet Interface

The **IO Module** communicates via an **Ethernet Interface** and is designed to interface seamlessly with other **MQTT-compatible devices**. The module acts as an **input/output device** that bridges the **MQTT network** and the connected gateway, enabling real-time data exchange over the MQTT protocol. This **Digital I/O module** is robust, energy-efficient, and user-friendly, making it ideal for a wide range of IoT applications. It supports for **digital inputs** and allows users to easily publish and subscribe to topics on an MQTT broker, facilitating communication with other devices, cloud platforms, and remote monitoring systems.

8 Port DO Module with MQTT Over Ethernet: -



The IO modules enclosures are DIN rail Mountable. It's a closed Enclosure with openings for connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors

2 Pin 5.08 mm pitch pluggable screw Terminal Block with max 2.5 sq mm wire connection.

Dimensions

110 mm B x 110 mm L x 50 mm H

Power

Input Power – 12 – 24 VDC or 24 V AC/DC

Typical – 12V DC @ 500mA

Operating Temperature

0 – 60° C (32 ~ 140°F)

Storage Temperature

-20 - 70° C (-4 ~ 158°F)

Storage Humidity

5 ~ 95 % RH, non – Condensing

Certifications



Relay Output –

Channels

8

Contact Form

1A, 1C

Contact Material

Ag Alloy

Contact Capacity

10A @ 240VAC, 10A @ 28VDC

Coil Voltage

5 – 48 VDC

Coil Power

0.36 W

Insulation Resistance

250MΩ

Electrical Life

1 x 10⁵

Mechanical Life

1 x 10⁷

Operating Time

7 msec, Max 15 msec

Release Time

2 msec, Max 6 msec

Technical Specifications:

- **MQTT Protocol Support:** Fully compliant with the MQTT protocol, supporting both MQTT 3.1.1 and MQTT 5.0.
- **Ethernet Connectivity:** Standard Ethernet port (10/100 Mbps) for stable and high-speed data transmission.
- **Low Power Consumption:** Optimized for low power usage, ideal for battery-operated or energy-efficient applications.
- **Easy Integration:** Supports standard libraries and tools for easy integration with existing systems.
- **Message Retention & Quality of Service (Q o S):** Supports various Q o S levels (0, 1, 2) to guarantee message delivery.
- **Secure Communication:** Support for TLS/SSL encryption for secure communication with MQTT brokers.
- **Real-time Data Handling:** Efficient, real-time messaging with minimal delay.
- **Compact Form Factor:** Small and lightweight, designed for embedded systems.

To set up MQTT over Ethernet and find a device's IP address using its MAC address, you'll follow these general steps. This involves setting up an MQTT broker (e.g., Mosquitto, MQTTX, or another broker), connecting devices to the network, and using the MAC address to find the device's IP. Below is a detailed procedure for setting up MQTT with Ethernet and using a MAC address to find the IP address of a device.

Configuration Center

Network Settings

IP Address:

Subnet Mask:

Gateway:

DNS Server:

MQTT Configuration

Host IP or Domain:

Topic:

Subscriber:

Username:

Password:

Port:

Data Rate/Sec:

(Image1.1)

Login and MQTT Configuration: -

1. Login Prompt

When you open the device's details page, you will be prompted to enter a **username** and **password**.

2. Default Username

The default **username** is usually admin.

3. Default Password

The default **password** is typically admin or password.

4. MQTT Port

The default MQTT port is **1883** for unencrypted connections.

For secure MQTT over TLS, use **port 8883**.

5. Topic Name

Ensure you configure the correct **topic name** where the analog and digital data will be published.

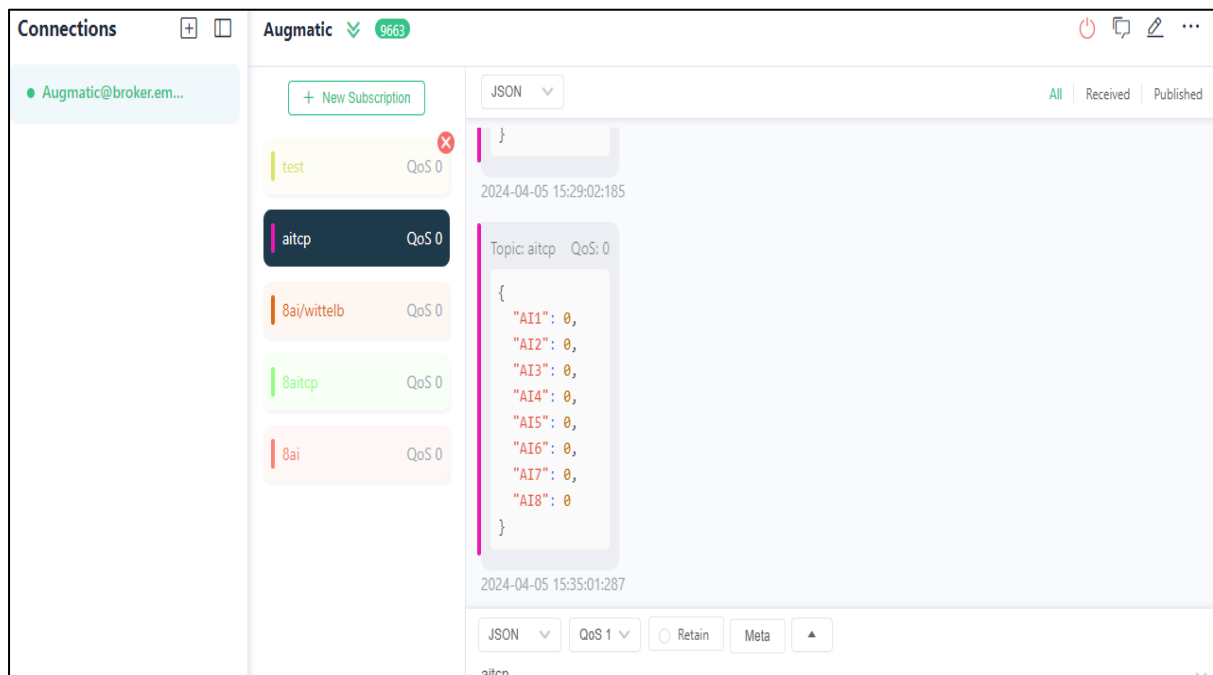
For example, if you set the topic to "aitcp", you will receive analog and digital data on that topic.

6. Data Rate

The **data rate** refers to how often the device publishes messages. You can set it to a specific interval, such as every second or every few seconds, depending on how often you want the data to update.

Note: (Username Password is optional).

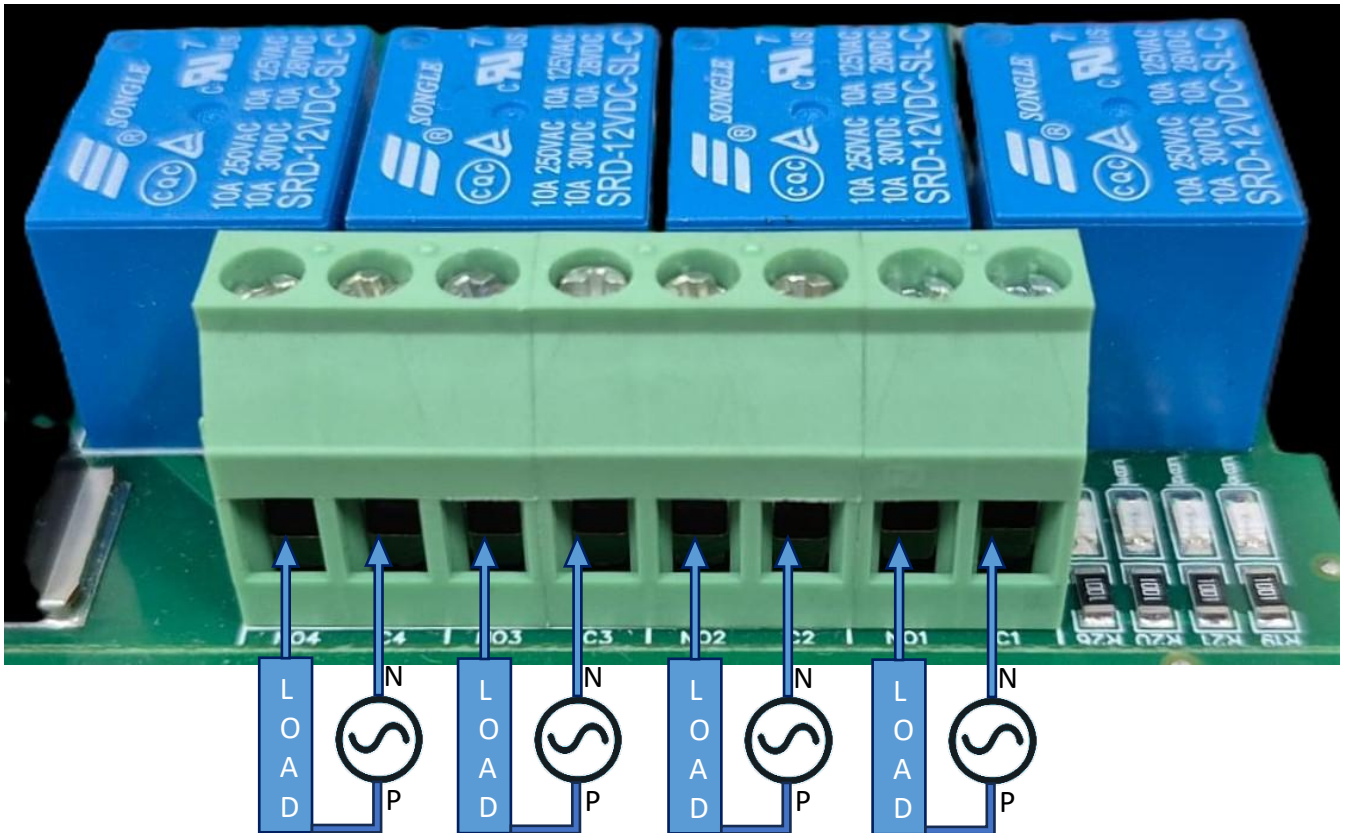
- **After submitting the data, you will get data on MQTT cloud.**



(Image1.2)

- Here you can see the data of digital output data of 8 channel.as shown in image1.2.

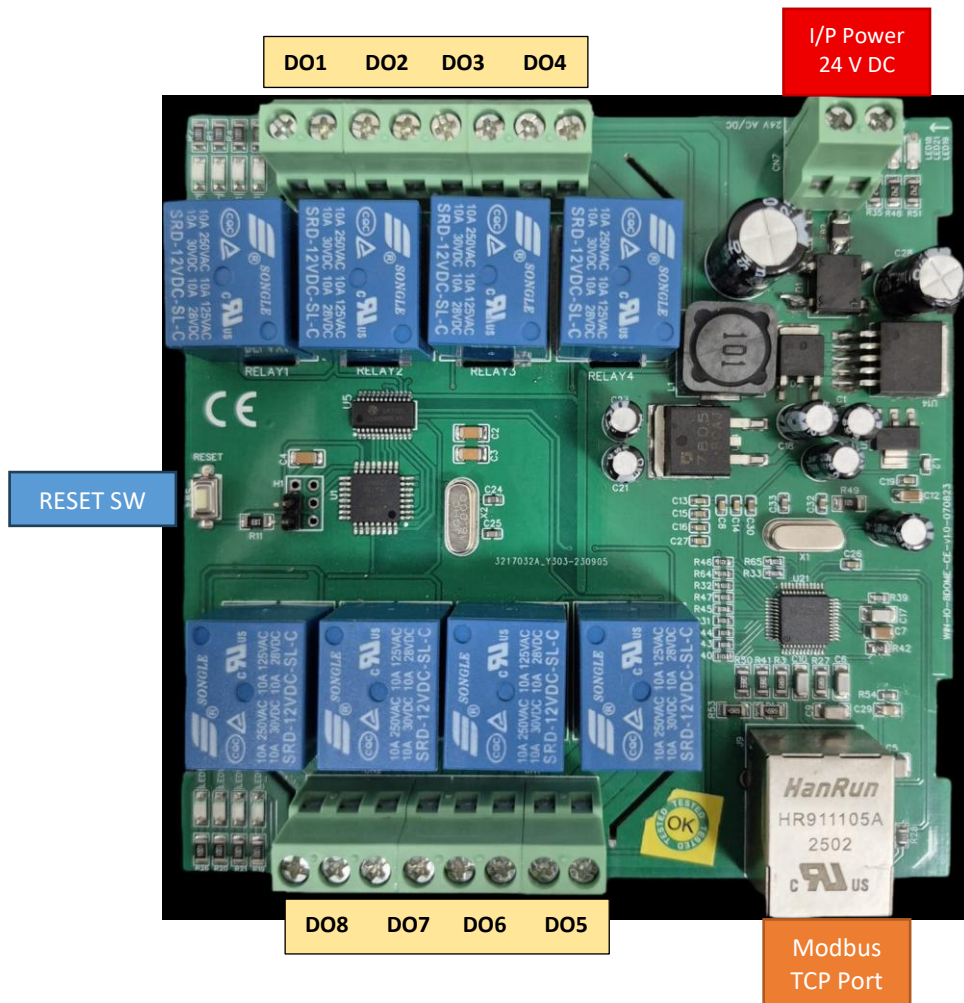
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C** and **NO** remains open, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

Module Details –



Note – Configuration Guide is separately provided for the configuration of the module.

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